End Semester Examination 2024

Name of the Course: Diploma
Name of the Paper: Data Structures

Semester: 4th
Paper Code: DTCS-404

Time: 3 Hours

Maximum Marks: 100

Note:

(i) All Questions are compulsory.
(ii) Answer any two sub questions among a, b and c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(10*2=20 marks)	
(a)	Define data structures and explain different types of data structure?	
(b)	Describe time and space complexities of an algorithm and explain the various notations used to denote it?	**CO1**
(c)	Discuss the characteristics of a good data structure and its significance in computer science?	
Q2	(10*2=20 marks)	**CO2**
(a)	(i) What is sorting? Write an algorithm for selection sort. (ii) Sort the following numbers using selection sort showing each step clearly. 4, 8, 3, 7, 13, 9, 25	
(b)	Explain quick sort algorithm with suitable example.	
(c)	Write a c code to perform Binary search on an array? Also find the space and time complexity.	
Q3	(10*2=20 marks)	
(a)	(i) Write the benefits of using a circular queue compared to a normal queue? (ii) Write a C code for ***insertion*** in a ***circular*** queue.	**CO3**
(b)	Write an algorithm for performing **PUSH** and **POP** operations on a Stack.	
(c)	Write an algorithm to determine if the parentheses in a given string are balanced. For example, "(()) and "(()()" are balanced, while "())(" and "())" are **not.**	
Q4	(10*2=20 marks)	
(a)	(i) What is a linked list and what are the benefits of using a linked list in comparison to an array? (ii) Write an algorithm or C function to insert a node at the end of a linked list.	**CO4**
(b)	Write an algorithm to insert a node in a BST and find time complexity for average and worst case.	
(c)	(i) Define binary search tree and explain its properties. (ii) Write an algorithm to search a given value in binary search tree.	

Q5	(10*2=20 marks)	CO5
(a)	Define a graph and discuss its different representations (adjacency matrix, adjacency list).	
(b)	Write an algorithm to perform Depth First Search(DFS) on a graph and how does it differ from Breadth First Search(BFS).	
(c)	(i) Write short on the concept of hashing and its benefits. (ii) What are the different ways to resolve the problem of collision in hashing?	